
Write the C++ program of the following problems.

Problem 1:

Write a C++ program to print the appropriate activity depending on the variable temperature and humidity value. The table below assumes that the temperature can only be warm and cold, and the humidity can only be dry and humid.

If temperature is	If humidity is	Print this activity
warm	dry	Play tennis
warm	humid	swim
cold	dry	Play basketball
cold	humid	Watch tv

Program 2:

Write a program that asks the user which country he/she lives in. If the country is not "**Germany**" or "**Australia"**, the program should print out "**You should come to visit these sometime!**".

Problem 3:

Jack is a teacher who needs a program that helps him to compile 8th class results. He has five subjects (English, Maths, Chemistry, Social Science, and Biology) marked in detail. Program asks the user to enter five subjects' marks, including student name and displays the total marks, percentage, grade (by percentage), and student name. Every subject has a total of 100 marks. Grading policy details are mentioned below

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Percentage	Grade
90-100 percentage	A+
80-90 percentage	A
70-80 percentage	B+
60-70 percentage	B
50-60 percentage	C
40-50 percentage	D
Below 40 percentage	F

The output should be like this
Student Name: Joe
Total Marks: 390
Obtain percentage: 78%
Obtained Grade: B+

Problem 4:

Write a Program to display an Astrological sign or a Zodiac sign for a given date of birth. Zodiac Signs according to the dates and months are given below.

Zodiac Sign	Dates of Birth	Symbol
Aries	Match 21 - April 19	The Ram
Taurus	April 20 - May 20	The Bull
Gemini	May 21 - June 20	The Twins
Cancer	June 21 - July 22	The Crab
Leo	July 23 - August 22	The Lion
Virgo	August 23 - September 22	The Virgin
Libra	September 23 - October 22	The Scales
Scorpio	October 23 - November 21	The Scorpion
Sagittarius	November 22 - December 21	The Archer
Capricorn	December 22 - January 19	The Goat
Aquarius	January 20 - February 18	The Water Bearer
Pisces	February 19 - Match 20	The Fishes

The program takes the date and month of birth as input, and then displays the astrological sign or Zodiac sign according to that input.

Test Cases

Input	Output	Explanation
Day = 10 Month = December	Sagittarius	People born on this data have zodiac Sagittarius.
Day = 7 Month = September	Virgo	

Problem 5:

Write a program that calculates and prints the bill for a cellular telephone company. The company offers two types of service: regular and premium. Its rates vary, depending on the type of service. The rates are computed as follows:

Regular service: $10.00 plus first 50 minutes are free. Charges for over 50 minutes are $0.20 per minute.

Premium service: $25.00 plus:

- For calls made during the day., the first 75 minutes are free; charges for more than 75 minutes are $0.10 per minute.
- For calls made during the night, the first 100 minutes are free; charges for more than 100 minutes are $0.05 per minute.

Your program should prompt the user to enter a service code (type char), and the number of minutes the service was used.

A service code of **r** or **R** means regular service; a service code of **p** or **P** means premium service. Treat any other character as an error. Your program should output the type of service, the number of minutes the telephone service was used, and the amount due from the user.

For the premium service, the customer may be using the service during the day and the night(**d** or **D** for day and **n** or **N** for night). Therefore, to calculate the bill, you must ask the user to input the number of minutes the service was used during the day and the number of minutes the service was used during the night.

Test Cases:

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Enter which type of customer you are(Residental or Premium) select (r/p) : p
(Press 'D' for day time or Press 'N' for night time) call : N
Enter number of mins you used the service : 57
The charges are : 25$
```

Problem 6:

A fruit shop during week days sells in the following prices:

Fruit	Price
banana	2.50
apple	1.20
orange	0.85
grapefruit	1.45
kiwi	2.70
pineapple	5.50
grapes	3.85

During the **weekend days** the prices are **higher**:

Fruit	Price
banana	2.70
apple	1.25
orange	0.90
grapefruit	1.60
kiwi	3.00

pineapple grapes	5.60 4.20

Write a program that **reads** from the console a **fruit** (banana / apple / …), **a day of the week** (Monday / Tuesday / …) and **a quantity (a decimal number)** and **calculates the price** according to the prices from the tables above. Print **"error"** if it is an **invalid day** of the week or an **invalid name** of a fruit.

Test Cases:

Input	Output
orange **Sunday** **3**	**2.70**
kiwi **Monday** **2.5**	**6.75**
grapes **Saturday** **0.5**	**2.10**
tomato **Monday** **0.5 error**	**tomato** **Monday** **0.5 error**

Problem 7:

A hotel offers **two types of rooms**: **studio and apartment**. Prices are in dollars ($). Write a program that calculates **the price of the whole stay for a studio and apartment**. **Prices** depend on the **month** of the stay:

May and October	June and September	July and August
Studio- 50$ / per night	Studio - 75$ / per night	Studio 76$ / per night
Apartment - 65$ / per night	Apartment - 68$ /per night	Apartment - 77$ / per night

The following **discounts** are also offered:

- For a **studio**, in case of **more than 7** stays in **May and October**: **5% discount**.
- For a **studio**, in case of **more than 14** stays in **May and October**: **30% discount**.
- For a **studio**, in case of **more than 14** stays in **June and September**: **20% discount**.

- For an **apartment**, in case of **more than 14** stays, **no limitation regarding the month: 10% discount**.

Input Data

The input data is read from the **console** and contains **exactly two lines**:

- The **first** line contains the **month** – **May**, **June**, **July**, **August**, **September** or **October**.
- The **second** line is the **number of stays** – **integer within the range** [**0 … 200**].

Output Data

Print the following **two lines** on the console:

- On the **first line**: "**Apartment: { price for the whole stay }$.**"
- On the **second line**: "**Studio: { price for the whole stay }$.**"

Test Cases:

Input	Output	Comments
May 15	Apartment: 877.50$. Studio: 525.00$.	In May, in the case of more than 14 stays, the discount for the studio is 30% (50 - 15 = 35), and for the apartment is 10% (65 - 6.5 = 68.5). The whole stay in the apartment: 877.50 lv The whole stay in the studio: 525.00 lv
June 14	Apartment: 961.80$. Studio: 1052.80$.	
August 20	Apartment: 1386.00$. Studio: 1520.00$.	

Problem 8:

1. A student has to attend **an exam at a particular time** (for example, at 9:30 am). They arrive in the exam room at a particular **time of arrival** (for example 9:40 am). It is considered that the student has arrived **on time** if they have arrived **at the time when the exam starts or up to half an hour earlier**. If the student has arrived **more than 30 minutes earlier**, the student has come **too early**. If they have arrived **after the time when the exam starts**, they are **late**.

Write a program that inputs the exam starting time and the time of student's arrival, and prints if the student has arrived **on time**, if they have arrived **early** or if they are **late**, as well as **how many hours or minutes** the student is early or late.

Input Data

Read the following **four integers** (one on each line) from the console:

- The first line contains **exam starting time (hours)** – an integer from 0 to 23.
- The second line contains **exam starting time (minutes)** – an integer from 0 to 59.
- The third line contains an hour **of arrival** – an integer from 0 to 23.
- The fourth line contains **minutes of arrival** – an integer from 0 to 59.

Output Data

Print the following on the first line on the console:

- "**Late**", if the student arrives **later** compared to the exam starting time.
- "**On time**", if the student arrives **exactly** at the exam starting time or up to 30 minutes earlier.
- "**Early**", if the student arrives more than 30 minutes **before** the exam's starting time.

If the student arrives with more than one minute difference compared to the exam starting time, print on the next line:

- "**mm minutes before the start**" for arriving less than an hour earlier.
- "**hh:mm hours before the start**" for arriving 1 hour or earlier. Always print minutes using 2 digits, for example "1:05".
- "**mm minutes after the start**" for arriving more than an hour late.
- "**hh:mm hours after the start**" for arriving late by 1 hour or more. Always print minutes using 2 digits, for example, "1:03".

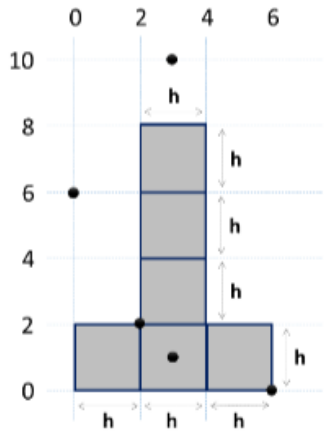
Test Cases

Input	Output
Exam Starting Time (hour): 9 Exam Starting Time (minutes): 30 Student hour of arrival: 9 Student minutes of arrival: 50	Late 20 minutes after the start
Exam Starting Time (hour): 16 Exam Starting Time (minutes): 0 Student hour of arrival: 15 Student minutes of arrival: 0	Early 1:0 hours before the start
Exam Starting Time (hour): 9 Exam Starting Time (minutes): 0 Student hour of arrival: 8 Student minutes of arrival: 30	On time 30 minutes before the start
Exam Starting Time (hour): 9 Exam Starting Time (minutes): 0 Student hour of arrival: 10 Student minutes of arrival: 30	Late 1:30 hours after the start
Exam Starting Time (hour): 14	On time

Exam Starting Time (minutes): 0 Student hour of arrival: 13 Student minutes of arrival: 55	5 minutes before the start
Exam Starting Time (hour): 11 Exam Starting Time (minutes): 30 Student hour of arrival: 8 Student minutes of arrival: 12	Early 3:18 hours before the start
Exam Starting Time (hour): 11 Exam Starting Time (minutes): 0 Student hour of arrival: 11 Student minutes of arrival: 0	On time
Exam Starting Time (hour): 11 Exam Starting Time (minutes): 30 Student hour of arrival: 10 Student minutes of arrival: 55	Early 35 minutes before the start
Exam Starting Time (hour): 11 Exam Starting Time (minutes): 30 Student hour of arrival: 12 Student minutes of arrival: 29	Late 59 minutes after the start

Problem 9:

1. The figure consists of **6 blocks with size h * h**, placed as in the figure below. The lower left angle of the building is at position {0, 0}. The upper right angle of the figure is on position {**2*h**, **4*h**}. The coordinates given in the figure are for **h = 2**:



Write a program that enters an integer **h** and the coordinates of a given **point {x, y}** (integers) and prints whether the point is inside the figure (**inside**), outside of the figure (**outside**) or on any of the borders of the figure (**border**).

Sample Input and Output:

Input	Output
2 3 10	Outside
2 3 1	Inside
2 2 2	Border
2 6 0	Border
2 0 6	Outside
15 13 55	Outside
15 29 37	Inside